10. Dataset Documentation

AI-Powered Energy Consumption Optimization Assistant

Dataset Name

Appliances Energy Prediction dataset.

Dataset Source

UCI Machine Learning Repository.

Dataset Link

https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction

Data Format

CSV (comma-separated values). The project uses **data/raw/energy_consumption.csv** as the raw dataset and **data/processed/energy_cases.csv** as the processed dataset.

Key Fields

Field	Description
date	Timestamp of each observation.
Appliances	Appliance energy consumption; the main consumption measure.
lights	Energy consumption associated with lights.
T1–T9 / RH_1–RH_9	Indoor temperature and relative-humidity measurements.
T_out / RH_out	Outdoor temperature and relative humidity.
Press_mm_hg, Windspeed, Visibility, Tdewpoint	Additional outdoor weather and environmental context.

Dataset Purpose

The dataset provides historical appliance-energy observations and environmental context for the assistant. The project converts observations into searchable energy cases for semantic and keyword retrieval, consumption-pattern analysis, and context-aware optimization recommendations. It supports analysis of high-consumption periods and rule-derived anomaly indicators; it does not itself contain verified records of savings achieved after applying recommendations.

Data Preprocessing Performed

- Parsed the date column into a timestamp.
- Derived time features such as hour, day of week, month, year, and season.
- Added time-period and peak-hour features, including period and is_peak_hour.
- Calculated rolling_mean_24 and consumption_deviation features.
- Derived consumption_level, anomaly_score, and is_anomaly using project processing rules.
- Generated case_id and case_description fields for retrieval and readable case context.
- Saved the processed records to data/processed/energy_cases.csv for ingestion and search.

Note: The project ingestion output reported 19,735 cases. The consumption categories and anomaly indicators are derived features, not independently verified ground-truth labels.